Qualcomm RB5 Flight Drone and OptiTrack Setup Manual

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1 Introduction

This manual has been made to help setup the Qualcomm RB5 Flight drone that is distributed and supplied by ModalAI. This drone is available in the Robotics lab D06, in the Colonel By Hall (CBY) building. This manual was created in the first quarter of 2024 and could be subject to changes as the ModalAI web-pages as well as any other links provided could have been updated/changed.

Before we start, it is important to note that the RB5 drone runs on Ubuntu 18.04 and it is necessary that the Operating System on the computer matches this version of Ubuntu (you can also use a Virtual Machine running Ubuntu 18.04). This drone will not be recognized on Windows/Mac OS. If ROS will be used to program the drone it is also imperative to note that due to the version of Ubuntu used by the drone, only ROS 1 (Melodic) will be compatible with it. The steps to download and setup ROS as well as MAVROS on the drone will be detailed in later sections.

2 Firmware Installation and Update

In order to install and update the drone firmware, connect the RB5 to the computer using a USB-C cable. To check if the drone is recognized by the computer, open a terminal and type ***'lsusb'*** and check wether Qualcomm appears in the list of devices connected.

Proceed to the following ModalAI software download page by first creating an account and logging in: RB5 Platform Releases. Download the latest firmware under the 'Protected Downloads' $->$ 'RB5 Flight with VOXL SDK Platform Release'. **Note:** Make sure to click on the VOXL SDK release and not RB5 Flight SDK release.

Once the installation is complete follow the steps listed on RB5 Upgrade Guide, including the 'backup files' steps as these files will be erased upon flashing.

2.1 Connecting RB5 to the Wifi

Firstly, RB5 needs to be connected to the wifi to perform the following tasks. To connect the drone to the wifi network, it can use the following two commands (use either one):

- `voxl wifi` or `nmcli dev wifi`
- When `voxl-wifi` is input, three options will be displayed, namely, `softap`, `staion` and `factory`:

  Softap: Broadcasts its own wifi to which other devices can connect to. The WiFi SSID and password need to be set by the user.

  Station: Connects to an external WiFi.

  Factory: Essentially the function is the same as softap, in this mode it resets it's WiFi SSID and password to the default settings.

- Select, station and enter the SSID (Wifi name) and the password when prompted.
- Once the connection is successful reboot the RB5 and when it is reconnected the WiFi IP address should be displayed in the boot up graphics on the terminal.
- If this method is not functioning as intended use the second method and type the following in the terminal:
  `nmcli dev wifi con [SSID] password [password]`

- Reboot and check for connection

2.2 Installing ROS

The RB5 drone has Ubuntu 18.04, therefore if you need to use ROS, download ROS Melodic as the higher versions will not be compatible (including ROS 2). The steps to install ROS Melodic first on your PC and then onto the drone are similar and quite simple. Therefore I will just be outlining the steps to install ROS on the drone. The steps are sourced from Modal AI's official website with slight alterations. Enter each of the following terminal commands one at a time, do not paste the entire block.

- To install ROS Melodic (as an example), connect RB5 Flight to the internet and run the following in sequence:

```
apt-get update
apt install -y ros-melodic-ros-base ros-melodic-image-transport
```

- If you have not already configured MPA and PX4, execute the following:

```
voxl-configure-mpa
voxl-configure-px4
```

  In both cases, select `no` when asked for factory default reset.

2.3 Setting Up PX4 Development Environment

The PX4 firmware will need to be installed using the following steps. This will not only run the SITL (Software in the loop) as well as HITL (Hardware in the loop) for simulation purposes but also includes all of the necessary files to make PX4 Autopilot run from an external computer.

- The setup steps can be followed using this official PX4 Youtube "Setting up PX4 Dev Environment" video. The steps from the video are listed here.

- Open a terminal and install git using:

```
sudo apt-get install git
```

- Clone the following PX4 repository:

```
git clone https://github.com/PX4/Firmware.git $--$recursive
```

- A git clone called "Firmware" should now be available in the "Home" directory.

- Now, enter the folder using the `cd Firmware` command in the same terminal.

- Type in the following script to build the firmware and enter the admin password when prompted:

```
bash ./Tools/setup/ubuntu.sh
```

- Next, reboot the system. Once it is back on, to verify that everything is working, open a terminal, navigate into the folder using the `cd` command, and type:

```
make px4\_sitl jmavsim
```

This will open the JMavSim simulator which will display a default drone along with the terminal from which commands can be sent.

- In the terminal type:

```
commander take off
```

If the drone takes off, everything is working fine and the simulation environment can now be used.

- If another simulator needs to be used, such as Gazebo, type in:

```
make px4\_sitl gazebo
```

3 Ground Control Station

When it comes to controlling drones from the ground, having the right software is key. One popular choice for drones running on ArduPilot is Mission Planner. It's known for its easy-to-use interface and a range of features like mission planning, real-time data display, and geofencing. It's a favorite among both casual users and professionals for its versatility.

For those using PX4 Autopilot, QGroundControl (QGC) is the go-to option. QGC is an open-source, comprehensive ground control station designed specifically to work seamlessly with PX4 systems, offering tools for mission planning, flight monitoring, and vehicle setup. One of QGC's standout features is its intuitive interface, which simplifies the process of mission planning and execution. Users can easily set up missions for navigation, define flight paths, and customize parameters to suit specific requirements. It provides real-time telemetry data display, allowing operators to monitor crucial flight metrics such as altitude, speed, and battery status.

As this guide is exclusively for a drone using a PX4 flight controller, QGC should be used.

To download QGC, visit the Github repository for QGroundControl and go the "Releases" tab. Here, download the version v4.2.0 which works with the version of Ubuntu (18.04) recommended in this Guide.

Note: Version discrepancies can occur and therefore, it is recommended to do some research if the Ubuntu version differs from the one mentioned here. Some higher versions of QGC require to be allowed through the Firewall settings to work properly.

3.1 Connecting to QGC

There are three ways to connect to the ground control station:

- USB connection (for calibration only)
- Drone Softap mode (over Wifi)
- Drone Station mode (over Wifi)

As explained in the earlier section, in the softap mode the drone broadcasts its own wifi to which the PC can connect to. Once this connection is established opening QGC will automatically connect it to the drone.

Although, this method is simple it cannot be used over a few meters as the connection is weak. It is therefore recommended to use the station mode.

The reason this connection is established is due to the VOXL MAVLink Server. One would notice that in station mode QGC does not recognize the drone directly (if the RB5 is factory new and not already configured to connect). In this situation follow the below steps:

- While the RB5 is connected using USB C, adb shell onto it and check IP address of the desktop running QGC (`ip address` in terminal).
- Make sure the VOXL MAVLink server is active by typing the following and checking the list:

```
voxl-inspect-services
```

- To access the MAVLink server config file, type in:

```
cat etc/modalai/voxl-mavlink-server.conf
```

- The `primary\_static\_gcs\_ip` should be kept at the default value of `192.168.8.10`. This is the softap/factory IP of the drone when it is broadcasting it's own WiFi, hence it is able to connect automatically.
- Change the `secondary\_static\_gcs\_ip` to the IP address of the desktop running QGC connected to the same external WiFi as the RB5.
- Press `Ctrl + O` to save and exit the config file. The drone should now automatically connect to QGC when it is connected on the same WiFi network.

3.2 Calibrating the Drone on QGC

Before the first flight of the drone it is crucial to check the calibration as well as if the drone model selected by the software matches the drone being used. It is also imperative to check if all four (2 CW and 2 CCW) rotors are assigned correctly. To do this follow the below steps:

- Click on the QGC icon on the top left when the app is open, select the option for "Vehicle Setup".
- First, under the "Airframe" tab, select the appropriate type of drone, the "Holybro S500" Airframe closely matches the RB5.
- Recalibrate all the sensors under the "Sensors" tab, starting with the Gyroscope. Follow the instructions on the screen which indicate how to place the drone. It is recommended to calibrate on the ground or a flat surface.
- Under the "Power" tab make sure the battery specifications are correctly input.
- The most important step is the "Actuators" tab, under which each of the propellers can be tested if they are functioning correctly and if they are rotating in the right direction. Make sure to assign the rotors to the correct numbers as per the diagram shown in the application.

4 OptiTrack System

OptiTrack motion capture system, well known for its robustness and precision, integrates a network of twenty one cameras, enveloping a rectangular netted cage within the laboratory. This setup offers a full-fledged 6 Degrees of Freedom (DOF) motion capture system, for acquiring comprehensive telemetry data. This motion capture system works with red security USB key and Motive, the companion software to this system that is installed on the University Lab computer. It is important to note that OptiTrack and Motive only work on Microsoft Windows operating system.

4.1 Calibrating the Cameras

To keep the camera system accurate and in optimal operative condition, frequent calibration is required. If the date of last calibration is more than two months old, re-calibration should be carried out.

- Plug in the power located at the entrance of the netted cage area (to the left when facing the cage entrance).
- Use the lab computer and insert the red USB key. Once the USB key starts flashing (which means it has established a successful connection), the cameras will have a green/blue ring around them
- Open Motive, open "Camera Preview" of all the cameras (this is done to visualize if all cameras receive wanding data during the calibration process).
- Make sure to check if the cameras are detecting any foreign bodies/trackers before starting. Any shiny elements will trigger a detection on the cameras and it is therefore recommended to keep any shiny items away during this process.
- To re-calibrate, click on "Begin Wanding" under the "Calibration" tab and use the wand provided with the system (should be available in the lab). This wand is typically a stick with reflective IR markers attached to it. These markers are used by the camera to locate the wand's position.
- Begin calibrating by entering the cage area and waving the wand in a figure eight pattern, slowly and carefully in front of each of the cameras present.
- Once thoroughly calibrated, i.e. when enough samples have been collected to cover each of the 2D camera previews edge to edge, click on "Calculate" and wait till the calibration is ready to apply.
- Once this is done, click on "Apply Result". The software will then ask to save the calibration file and set the ground plane (usually you do not need to change the value for the ground plane). Once saved, proceed by clicking 'Set Ground Plane' keeping the default value.
- The calibration should now be complete.

4.2 Creating a Rigid Body

To use the software to track the drone, a rigid body needs to be created using reflective IR markers attached to it. This step is crucial as motion capture uses this rigid body for measurements as well as any visualization going forward. If different drones are used, it is advised to create a rigid body for each of the separate drones. Below are the steps detailing how to execute this process.

- Attach the markers to the drone using a double-sided tape (5 - 6 markers). Make sure majority of them are attached at relatively different heights for better accuracy.
- Start the camera system and place the drone in the center of the cage. The center is the origin point

of the two axes drawn on the ground.

- The markers should be visible on the camera preview (2D or 3D). Select the markers by dragging the mouse across them while holding the left mouse button.

- While the markers are selected, right click and choose `Rigid Body>Create From Selected Markers`.

- Rename the rigid body under the "Assets" tab. The rigid body is now saved and can be activated by ticking the box in the "Assets" tab next to its name.

- On the next restart of Motive, if the rigid body is not automatically displayed when the drone is placed in the same spot, it can be activated as explained above.

- Make sure to always place the drone in the same spot and orientation in the cage.